

Sayantana Mukhopadhyay

Curriculum Vitae

Senior Research Scholar (SRF)

National Institute of Science Education and Research

(NISER), India

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About Me

I am a Senior Research Scholar at the National Institute of Science Education and Research (NISER), Bhubaneswar, working in the field of main-group organometallic chemistry. During my doctoral research under the guidance of Prof. Sharanappa Nembenna, I have developed broad expertise in the **design, synthesis and reactivity of metal complexes**, with a particular focus on **catalytic transformations and mechanistic investigations**. My research encompasses **organometallic synthesis, catalyst development, small-molecule activation, hydroboration, C-C bond formation and CO₂ activation**, supported by detailed kinetic, spectroscopic and structural studies. I am interested in **developing new organometallic systems, catalytic methodologies and mechanistic approaches**, and in exploring their applications across inorganic and organometallic chemistry.

Technical Expertise

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| • Air/moisture-sensitive reactions | • NMR / UV-VIS / IR Spectroscopy |
| • Glovebox Operation | • Mass Spectrometry |
| • Schlenk line Operation | • Single crystal XRD |
| • Chromatography (TLC/CC) | • OLEX 2.0 Software |
| • Crystallization | • Endnote X9, Diamond 4.6 Software |
| • Kinetic Study, Isotope labeling | • Chemdraw 16.0, Mast Renova 14.0 |

Education

BSc Chemistry First Class (H)

2014–2017

Vidyasagar University

MSc Chemistry First Class

2017–2019

Chhatrapati Sahuji Maharaj University

Certification

Graduate Aptitude test in Engineering 2020 (GATE score 460)

PhD Thesis

Title: Bis-Guanidinate-Supported Magnesium Complexes Synthesis, Reactivity, Catalytic Applications

Supervisor: Prof. Sharanappa Nembenna

Time Period: 2021-2026 (Ongoing)

Key Research Contributions:

- Development of bis-guanidinate magnesium (II) amide, and hydride complexes
- Reduction of carbodiimides and cyanamides using bis-guanidinate Mg hydride catalyst
- Substrate-assisted hydroboration of nitriles and isocyanides using Mg amide catalyst
- Development of bis-guanidinate magnesium chalcogenide chemistry (yet to publish)

Languages

Bengali, Hindi, English

Conferences & Awards

- *Main Group Molecules to Materials (MMM-IV)*, 2025 — Poster Presentation (**Best Poster Award**)
- *International Conference on Main Group Synthesis and Catalysis (ICMGSC-II)*, 2025 — Poster Presentation
- *International Conference on Main Group Synthesis and Catalysis (ICMGSC-I)*, 2023 — Poster Presentation
- *International Conference on Modern Trends in Inorganic Chemistry (MTIC-XIX)*, 2022 — Poster Presentation
- *Main Group Molecules to Materials (MMM-II)*, 2021 — Attended
- *DAE Conclave*, 2024 — Poster Presentation

Publications

Journal Articles

1. **Mukhopadhyay, S.**; Rajput, S.; Sahoo, R. K.; Nembenna, S., Insights Into the Magnesium-Catalyzed C–C Coupling of Terminal Alkynes with Carbodiimides. *Eur. J. Org. Chem.* **2024**, 27 (44), e202400702.
2. **Mukhopadhyay, S.**; Padhan, S.; Das, A.; Nembenna, S., Bis-guanidinate magnesium amide-catalyzed hydroboration of nitriles and isocyanides via a substrate-assisted pathway. *Org. Chem. Front.* **2026**, 13 (2), 350-363.
3. **Mukhopadhyay, S.**; Das, A.; Padhan, S.; Nembenna, S., Bulky bis-guanidinate ligand-supported magnesium hydride: Synthesis, reactivity, and catalytic applications. *Inorg. Chim. Acta.* **2026**, 589, 122924.
4. **Mukhopadhyay, S.**; Das, A.; Rajput, S.; Behera, G.; Nembenna, S., Mechanistic Studies of Bis(Guanidinate) Magnesium Hydride as a Precatalyst for the Hydroboration of Organic Carbonates and Carbon Dioxide. *Catal. Sci. Technol.* (2026), 10.1039/D6CY00898D.
5. **Mukhopadhyay, S.**; Das, A.; Rajput, S.; Prajapati, D. J.; Nembenna, S., Reactivity of Bis(Guanidinate) Magnesium Hydride and Alkyl Complexes: Mg-E (E = S, Se, Te) Bond Formation and Structural Diversification. *Inorg. Chem.* **2026**, 65, 20064–20077.
6. **Mukhopadhyay, S.**; Sahoo, R. K.; Patro, A. G.; Khuntia, A. P.; Nembenna, S., Low-valent germanium and tin hydrides as catalysts for hydroboration, hydrodeoxygenation (HDO), and hydrodesulfurization (HDS) of heterocumulenes. *Dalton Trans.* **2024**, 53 (45), 18207-18216.
7. Sarkar, N.; Sahoo, R. K.; **Mukhopadhyay, S.**; Nembenna, S., Organoaluminum Cation Catalyzed Selective Hydrosilylation of Carbonyls, Alkenes, and Alkynes. *Eur. J. Inorg. Chem.* **2022**, 2022 (8), e202101030.
8. Das, A.; **Mukhopadhyay, S.**; Rajput, S.; Nembenna, S., Magnesium catalyzed hydroamination of carbodiimides and hydroboration of cyanamides. *Dalton Trans.* **2025**, 54 (47), 17553-17563.
9. Sahoo, R. K.; **Mukhopadhyay, S.**; Rajput, S.; Jana, A.; Nembenna, S., Bulky bis-guanidine ligand-based neutral and cationic zinc alkyl, halide, and hydride complexes: synthesis, characterization, and catalytic application. *Dalton Trans.* **2025**, 54 (41), 15554-15560.

Publications

10. Das, A.; **Mukhopadhyay, S.**; Nembenna, S., Conversion of Alkynes to Imidazolidin-2-ones: Insight into the Role of Magnesium Amido Complexes in the Reaction. *Organometallics* **2026**, *45* (1), 36-47.
11. Das, A.; **Mukhopadhyay, S.**; Rajput, S.; Nembenna, S., Catalytic hydroboration of α,β -unsaturated ketones using β -diketiminate magnesium hydride and mechanistic insights. *Dalton Trans.* **2026**, *55* (15), 6056-6066.
12. Padhan, S.; Rajput, S.; **Mukhopadhyay, S.**; Sabharwal, G.; Nembenna, S., Bis-Guanidinate Aluminum Complexes: Synthesis and Catalytic Application in Three-Component Aminophosphorylation of Aldehydes. *Organometallics* **2026**, *45* (12), 1384-1394.
13. Rajput, S.; Padhan, S.; Thilakan, N. M.; **Mukhopadhyay, S.**; Nembenna, S., Nickel-catalyzed dehydrogenative Zn-Zn coupling to a Zn(I) dimer and its reactivity. *Chem. Commun.* **2026**, *62* (2), 534-538.
14. Thilakan, N. M.; Rajput, S.; **Mukhopadhyay, S.**; Nembenna, S., Zinc-Mediated C-H Borylation of Terminal Alkynes: Mechanistic Insights. *Asian J. Org. Chem.* **2026**, *15* (3), e00637.

Review Article

15. **Mukhopadhyay, S.**; Patro, A. G.; Vadavi, R. S.; Nembenna, S., Coordination Chemistry of Main Group Metals with Organic Isocyanides. *Eur. J. Inorg. Chem.* **2022**, *2022* (31), e202200469.

Book Chapters

16. Nembenna, S.; Sarkar, N.; Sahoo, R. K.; **Mukhopadhyay, S.**, 2.03 - Organometallic Complexes of the Alkaline Earth Metals. In *Comprehensive Organometallic Chemistry IV*, Parkin, G.; Meyer, K.; O'hare, D., Eds. Elsevier: Oxford, 2022; pp 71-241.
17. **Mukhopadhyay, S.**; Rajput, S.; Nembenna, S., Guanidinate Ligand Stabilized Alkaline Earth Metal Complexes, Synthesis, Reactivity, and Catalytic Applications. In *Encycl. Inorg. Bioinorg. Chem.*, 2023; pp 1-21.
18. Prajapati, D. J.; Das, A.; **Mukhopadhyay, S.**; Nembenna, S., Zinc-Catalyzed Hydrofunctionalization Reactions. In *Encycl. Inorg. Bioinorg. Chem.*, 2026; pp 1-50.

Referee

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